

Enhancing Flow-Based Protein Structure Generation with Multi-Modal Protein Encoders

Vladimir Radenkovic
vr375@cam.ac.uk

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1 Introduction and Motivation

De novo protein design, the creation of novel proteins with specified functions and properties, represents a grand challenge in molecular biology and biotechnology with the potential to revolutionize medicine, industrial catalysis, and material science. While deep generative models have emerged as a powerful paradigm for this task, their training comes with a significant computational cost. A significant portion of this cost, as identified in analogous domains like image generation, stems from a fundamental "representation bottleneck" [8]. During training, the generative model are burdened with simultaneously learning two distinct and difficult tasks: 1) learning a high-quality, semantically meaningful internal representation of the complex data and 2) learning the complex, often pixel-level (or in this case, coordinate-level) process of reconstructing the data from this representation.

The denoising or flow-matching objective alone may be an insufficient signal for efficient representation learning, leading to slow convergence and requiring massive datasets and computational budgets [8, 7].

This proposal is founded on central hypothesis: **we can drastically accelerate the training and improve the output quality of a state-of-the-art protein backbone generator, Proteina [3], by aligning its internal representations with those derived from powerful pre-trained protein encoders.** To achieve this, **we will adapt Representation Alignment (REPA) [8],** a technique proven to improve both training efficiency and generation quality of diffusion transformers, to protein domain. We will use rich, multi-modal representations captured by specialised protein encoders to make training more efficient and improve generation quality, by sampling proteins with superior physical realism and high "designability" and diversity.

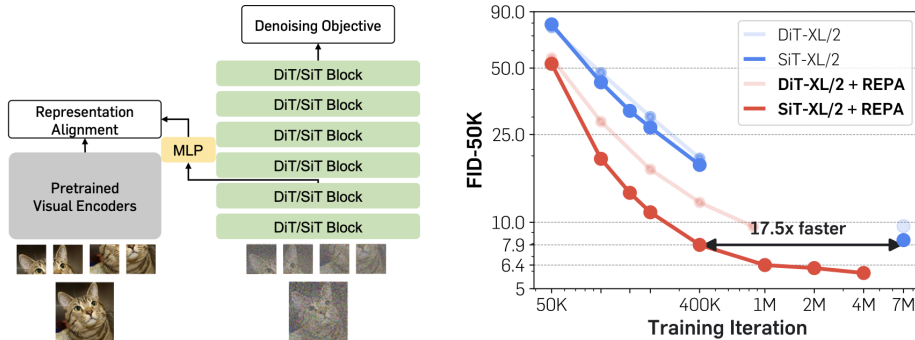


Figure 1: **Representation alignment(REPA)** aligns the diffusion model’s internal representations with powerful pretrained visual representation through a simple regularization. The model training becomes significantly more efficient and effective.

2 Foundational Methods and Models

- **Proteina [3]:** A highly scalable flow-based generative model for protein backbone. It’s non-equivariant transformer architecture by recent advances in vision-based diffusion transformers. The model achieves state-of-the-art result and enables generation of long and diverse protein chains. Proteina learns a continuous flow that transforms a simple Gaussian distribution into the complex distribution of realistic protein structures and can be conditioned on hierarchical fold class labels.

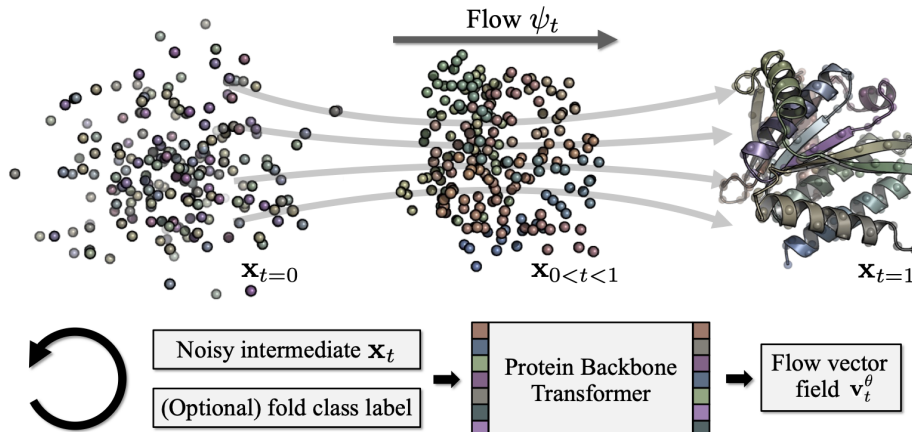


Figure 2: **Proteina Architecture.** The model uses a flow-matching to learn a flow to transform a Gaussian distribution over initial protein backbone’s C_α coordinates into realistic protein structures. Its diffusion transformer architecture allows for conditioning on hierarchical fold class labels.

- **Representation Alignment (REPA) [8]:** A regularisation technique that forces diffusion model’s internal representations to match the representations of a powerful pre-trained external encoder. This relieves the generator from learning semantic features from scratch, allowing it to focus on the generation task. During training, a standard flow-matching objective is augmented with an alignment loss term L_{REPA} that aligns hidden states of generative models on noisy inputs, with the representation of the corresponding clean data point as computed by a pretrained external encoder model. REPA has been shown to accelerate the training of diffusion transformers for image generation by over 17.5x.

3 Proposed Research Direction

Our strategy is to enrich the training of Proteina by aligning its internal representations with features from possibly three distinct and complementary classes of pretrained protein encoders.

1. **Protein Language Models (pLMs):** we can align Proteina’s internal states with embeddings from established pLMs to de-still evolutionary and structural knowledge.
 - **ESM2 [5]:** Aligning with ESM2 will guide Proteina to generate backbones that are consistent with the structural patterns implicitly learned from evolutionarily related protein sequences.
 - **ProstT5/Saprot [4, 6]:** These models are trained on joint sequence and structure information, using a discretized structural alphabet (3Di tokens). This alignment will provide a more direct and explicit structural by enforcing realistic local geometries.

2. **Machine-Learned Interatomic Potentials (MLIPs):** To enforce physical realism, we will align Proteina’s representations with the per-atom feature vectors from the state-of-the-art pretrained MLIP.
 - **MACE [1]:** By aligning with the equivariant representations of MACE, we can directly embed principles of physics and equivariance into the generative process. This will promote generation of structures with correct bond lengths, angles and stereochemistry, potentially reducing the need for extensive data augmentation during training.
3. **Inverse Folding Models:** By aligning Proteina’s latent states with backbone representations of inverse folding models, we will directly optimize for designability, the ability of a backbone to accommodate a stable amino acid sequence.
 - **ProteinMPNN [2]:** Aligning with internal backbone representations of ProteinMPNN will explicitly guide Proteina toward generating structures that are not just physically plausible but also highly "designable".

References

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